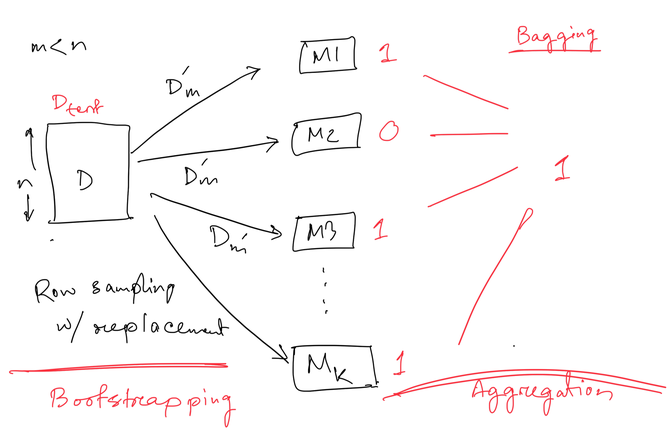
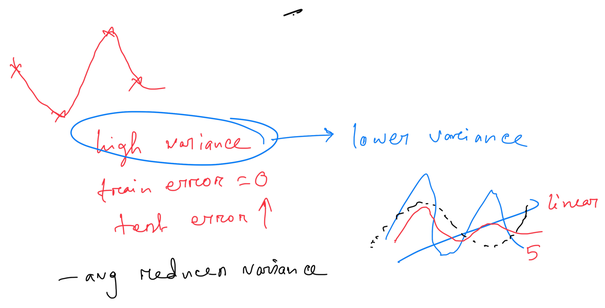
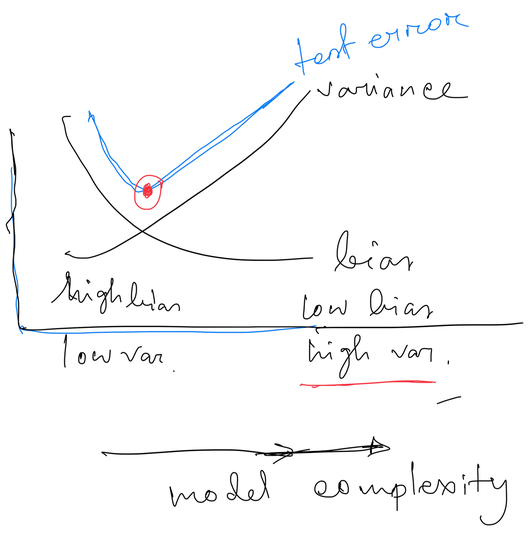
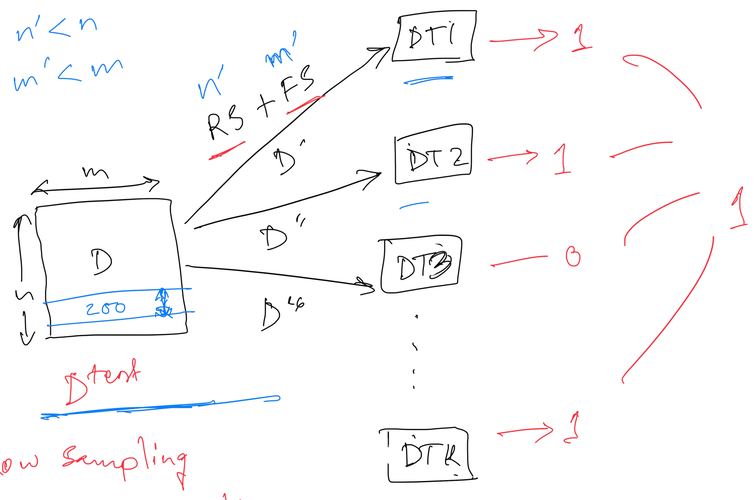






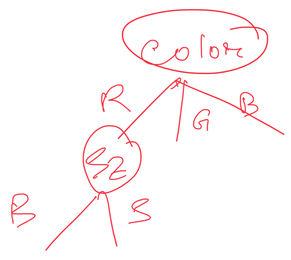


Color, sz, height, age

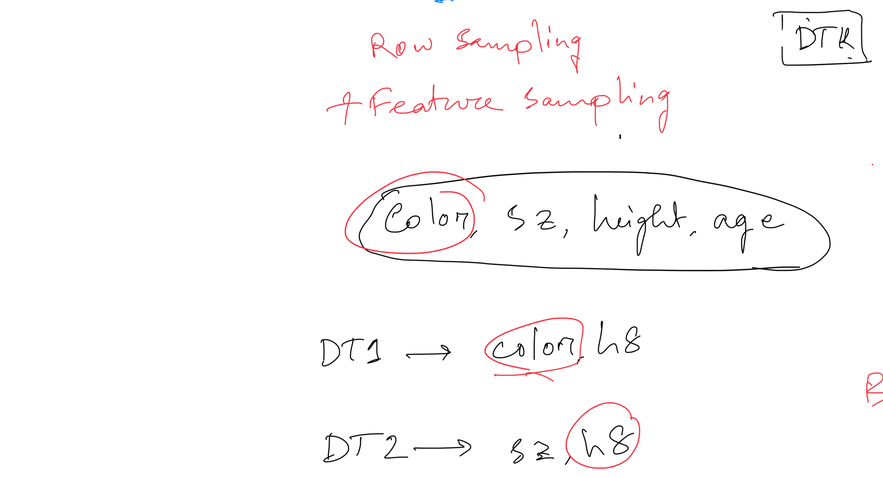
DT1 → color, h8

DT2 → sz, h8

DT3 → color, sz



Boosting – seq. model
– weak base learners



$$h(x) = \frac{\sum w_i h_i(x)}{n}$$

Adaboosting

pick a random no.

0.43 0.05 0.51

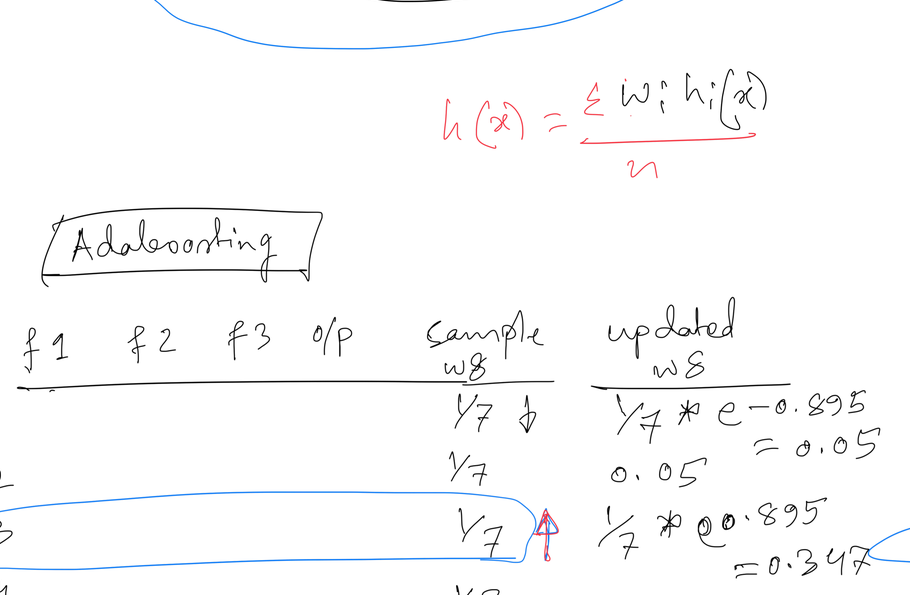
s3 s1 s3

	f1	f2	f3	o/p	sample w8	updated w8	Norm. w8
s1					1/7 ↓	1/7 * e^-0.895 = 0.05	0.07
s2					1/7	0.05	0.07
s3					1/7	1/7 * e^0.895 = 0.347	0.513
s4					1/7	0.05	0.07
s5					1/7	0.05	0.07
s6					1/7	0.05	0.07
s7					1/7	0.05	0.07

0-0.07 | 0.07 - 0.58

$D(i) = \frac{1}{n} = \frac{1}{7}$

1 ?



(*) Total error, $\epsilon_t = \frac{1}{7}$ 2 error → 2/7, 3 → 3/7

(*) Perf of stump, α_i

$= \frac{1}{2} \log_2 \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$

$= \frac{1}{2} \log_2 \left(\frac{1 - 1/7}{1/7} \right) = 0.895$

(*) Update weight

↑ error n → old w8 * e^{α_i}

↓ correct example weights → old w8 * $e^{-\alpha_i}$

$\sum 1 \times 0.6 + 0 \times 0.4 + 1 \times 0.7 =$ □